

Knowledge Test

Question 1: Fundamental Concepts

Which of the following statements correctly describes the difference between *IBNR* and *IBNER*?

- A. IBNR refers to claims that have been reported but remain unpaid; IBNER refers to claims that have already been paid but are claimed again by the policyholder.
- B. IBNR (*Incurred But Not Reported*) refers to claims that have occurred but have not yet been reported to the insurer; IBNER (*Incurred But Not Enough Reserved*) refers to reported claims whose individual reserves are estimated to be insufficient.
- C. IBNR and IBNER are equivalent and interchangeable concepts in international actuarial practice.
- D. IBNER applies exclusively to life insurance; IBNR applies only to property and liability insurance.

Answer B

IBNR (*Incurred But Not Reported*): claims that have already occurred during the coverage period but are not yet known to the insurer, either because the policyholder has not reported them or because the reporting process involves an inherent delay. This is the most uncertain component of the claims reserve.

IBNER (*Incurred But Not Enough Reserved*): claims that have already been reported, but whose individual reserves (RBNS – *Reported But Not Settled*) are estimated to be insufficient to cover the final settlement. This may result from adverse claims development or legal/medical changes after notification.

In practice, the **total IBNR reserve** estimated by the actuary usually incorporates both concepts: pure IBNR (unreported claims) and IBNER (adverse development of reported claims). In triangle-based methods, they are estimated jointly from the projected *ultimate*.

Options A, C, and D are incorrect: A confuses IBNR with RBNS; C is false (they are distinct concepts); D imposes a restriction by line of business that does not exist in standard actuarial theory.

Exercise 2: Calculation — Chain Ladder

Using the following table of cumulative paid claims (COP millions), calculate the volume-weighted average development factors and estimate the total IBNR reserve for 2022 and 2023.

Accident year	12 months	24 months	36 months (ult.)
2021	100	150	165
2022	120	180	?
2023	140	?	?

Formula: volume-weighted development factor

$$f_k = \frac{\sum_i C_{i,k+1}}{\sum_i C_{i,k}}.$$

Answer

Step 1: Volume-weighted development factors

$$f_{12 \rightarrow 24} = \frac{C_{2021,24} + C_{2022,24}}{C_{2021,12} + C_{2022,12}} = \frac{150 + 180}{100 + 120} = \frac{330}{220} = 1.5000,$$
$$f_{24 \rightarrow 36} = \frac{C_{2021,36}}{C_{2021,24}} = \frac{165}{150} = 1.1000.$$

Step 2: Projection to *ultimate*

All amounts below are expressed in COP millions.

For 2022, which already has 24 months of development:

$$\begin{aligned}\text{Ultimate}_{2022} &= 180 \times 1.1000 = 198.0, \\ \text{IBNR}_{2022} &= 198.0 - 180.0 = 18.0.\end{aligned}$$

For 2023, which has only 12 months of development:

$$\begin{aligned}\hat{C}_{2023,24} &= 140 \times 1.5000 = 210.0, \\ \text{Ultimate}_{2023} &= 210.0 \times 1.1000 = 231.0, \\ \text{IBNR}_{2023} &= 231.0 - 140.0 = 91.0.\end{aligned}$$

Therefore,

$\text{IBNR}_{\text{total}} = 18.0 + 91.0 = 109.0 \text{ million COP.}$

Cumulative development factors (CDF, *age-to-ultimate*)

$$\text{CDF}_{2022} = 1.10,$$

$$\text{CDF}_{2023} = 1.10 \times 1.50 = 1.65.$$

The year 2023 accounts for approximately 83% of total IBNR (91 out of 109), reflecting its high uncertainty due to limited development: a key argument for applying Bornhuetter–Ferguson to immature years.

Exercise 3: Multiple Choice — Bornhuetter–Ferguson

Under the Bornhuetter–Ferguson method, the IBNR reserve is calculated as follows:

A.

$$\text{IBNR}_{\text{BF}} = \text{Reported claims} \times (\text{CDF} - 1).$$

B.

$$\text{IBNR}_{\text{BF}} = q \times \text{ELR} \times \text{Earned premium},$$

where $q = 1 - 1/\text{CDF}$ is the percentage of risk that has not yet developed.

C.

$$\text{IBNR}_{\text{BF}} = \frac{\text{Ultimate}_{\text{CL}} + \text{Ultimate}_{\text{a priori}}}{2}.$$

D.

$$\text{IBNR}_{\text{BF}} = \text{Earned premium} \times \text{Historical loss ratio}.$$

Answer B

The formula for the Bornhuetter–Ferguson method is:

$$\text{IBNR}_{\text{BF}} = q \times M_{\text{a priori}}.$$

Here,

$$q = 1 - \frac{1}{\text{CDF}},$$

$$M_{\text{a priori}} = \text{ELR} \times \text{Earned premium},$$

where q is the percentage of risk that has yet to develop, $M_{\text{a priori}}$ represents expected claims under the a priori estimate, and CDF is the cumulative factor to *ultimate*.

The BF *ultimate* combines emerging experience with the a priori estimate:

$$\text{Ultimate}_{\text{BF}} = C_{\text{reported}} + q \times M_{\text{a priori}}.$$

Key concept: BF weights observed experience in proportion to how much the year has developed ($1-q$), and the a priori estimate in proportion to the remaining development (q). For very immature years ($q \rightarrow 1$), the result is almost identical to the a priori estimate. For mature years ($q \rightarrow 0$), the result approaches Chain Ladder.

Option A is the Chain Ladder formula, not the BF formula. Options C and D are incorrect.

Exercise 4: Calculation — Chain Ladder vs. Bornhuetter–Ferguson

The following information is available for accident year 2023:

- Cumulative paid claims to date: \$750,000.
- *Age-to-ultimate* CDF: 1.40.
- Earned premium: \$2,500,000.
- A priori ELR: 55%.

- (a) Calculate IBNR using Chain Ladder.
- (b) Calculate IBNR using Bornhuetter–Ferguson.
- (c) Under what circumstances would you choose BF over Chain Ladder? List at least three.

Answer

(a) Chain Ladder

$$\begin{aligned}\text{Ultimate}_{\text{CL}} &= 750,000 \times 1.40 = 1,050,000, \\ \text{IBNR}_{\text{CL}} &= 1,050,000 - 750,000 = \boxed{\$300,000}.\end{aligned}$$

(b) Bornhuetter–Ferguson

$$\begin{aligned}q &= 1 - \frac{1}{1.40} \approx 0.2857 \quad (28.57\% \text{ remaining}), \\ M_{\text{a priori}} &= 55\% \times 2,500,000 = 1,375,000, \\ \text{IBNR}_{\text{BF}} &= \left(1 - \frac{1}{1.40}\right) \times 1,375,000 \approx \boxed{\$92,857}.\end{aligned}$$

BF estimates a reserve approximately \$92,857 higher than CL. This indicates that cumulative paid claims (\$750,000) are relatively low compared with the a priori estimate (\$1,375,000 expected, with

28.57% of development remaining), and BF adjusts the estimate upward by giving more weight to the portfolio's expected behavior.

(c) I would prefer BF over Chain Ladder when:

- The accident year has **limited development** (high q , close to 1), and the data are sparse or unstable.
- The portfolio has experienced **significant changes** in business mix, volume, or underwriting, making historical extrapolation unreliable.
- Reported claims are **abnormally low or high** due to unusual factors (administrative delays, catastrophic events) that do not reflect expected behavior.
- There is a **sound basis for the a priori estimate**: rigorous pricing, market studies, or industry benchmarks.
- The triangle is **short** (few years of history) or **staggered** (missing data), limiting the stability of Chain Ladder.

Exercise 5: Multiple Choice — Mack Model

Mack's stochastic model (1993) provides a probabilistic foundation for the Chain Ladder method. Which of the following correctly describes its assumptions?

- A. Cumulative claims for each year follow a Poisson distribution, and development factors are constant across all periods.
- B.

$$\begin{aligned}\mathbb{E}[C_{i,j+1} \mid C_{i,j}] &= f_j \cdot C_{i,j}, \\ \text{Var}[C_{i,j+1} \mid C_{i,j}] &= \sigma_j^2 \cdot C_{i,j},\end{aligned}$$

and accident years are independent of one another.

- C. Claims development follows a continuous-time Markov process with constant intensity λ within each period.
- D. Development factors are lognormally distributed random variables, independent across periods and accident years.

Note: Mack's model allows the *Mean Squared Error of Prediction* of Chain Ladder to be quantified without assuming a specific distribution.

Answer B

Mack's stochastic model (1993) — *Distribution-free calculation of the standard error of chain ladder reserve estimates* — establishes three fundamental assumptions:

H1: The development factor determines the conditional expectation:

$$\mathbb{E}[C_{i,j+1} \mid C_{i,1}, \dots, C_{i,j}] = f_j \cdot C_{i,j}.$$

H2: The variance is proportional to the claims volume:

$$\text{Var}[C_{i,j+1} \mid C_{i,j}] = \sigma_j^2 \cdot C_{i,j}.$$

H3: Accident years are **independent** of one another.

The strength of Mack's model is that it **does not require a specific distribution** for claims: it only requires the first two moments. Under these assumptions, the Chain Ladder estimator of f_j is unbiased, and Mack derived closed-form formulas for the reserve's *Mean Squared Error of Prediction* (MSEP):

$$\text{MSEP}(\hat{R}_i) = \hat{C}_{i,n}^2 \sum_{j=k}^{n-1} \left[\frac{\sigma_j^2}{\hat{f}_j^2} \left(\frac{1}{C_{i,j}} + \frac{1}{\sum_{\ell} C_{\ell,j}} \right) \right].$$

This provides the basis for quantifying reserve uncertainty without resorting to *bootstrap* or simulation, a key requirement for Solvency II and high-quality actuarial reporting.

Exercise 6: Multiple Choice — Colombian Insurance Regulation

Under Decree 2555 of 2010 and the circulars issued by the Financial Superintendency of Colombia (SFC), which technical reserves are mandatory for insurers authorized to operate in non-life lines of business?

- A. Only the Unearned Premium Reserve (RPND) and the Reserve for Claims Pending Settlement.
- B. The Unexpired Risk Reserve (RRC), the Reserve for Reported Claims Pending Settlement (RSAPL), the IBNR Reserve, and the Catastrophe Reserve for lines of business that require it.
- C. Only the Mathematical Reserve calculated using current mortality tables approved by the SFC.
- D. An Overall Technical Reserve calculated as 12% of the earned premium for the accounting period.

Answer B

The Colombian regulatory framework for technical reserves (Decree 2555 of 2010, Part 2, Book 31, Title 1, and Chapter III of the SFC's Basic Legal Circular) establishes the following mandatory reserves for non-life lines of business:

- **Unexpired Risk Reserve (RRC):** covers the unexpired portion of the risk at the end of the period. The most common method is *pro rata temporis* (twentieths for monthly policies).
- **Reserve for Reported Claims Pending Settlement (RSAPL):** an individual or statistical estimate for each known but unpaid claim.
- **IBNR Reserve:** claims incurred but not reported, calculated using actuarial methods (CL, BF, etc.).
- **Catastrophe Reserve:** mandatory for lines exposed to catastrophic events (earthquake, compulsory motor bodily injury insurance (SOAT), terrorism). It accumulates in good years and is released when catastrophic claims occur.

For life insurance lines, the **Mathematical Reserve** is also required (the actuarial present value of future obligations less future premiums). Option D does not exist in Colombian regulation; fixed percentages are incompatible with the SFC's actuarial philosophy.

Exercise 7: Multiple Choice — Cape Cod Method

The Cape Cod method (Stanard–Bühlmann) differs from classical Bornhuetter–Ferguson primarily because:

- A. It uses only the three most recent accident years, discarding the older years in the triangle.
- B. The a priori ELR is not set externally: it is estimated directly from the portfolio data:

$$\text{ELR}_{\text{CC}} = \frac{\sum_i C_{i,\text{dev.}}}{\sum_i (P_i \times \% \text{ developed}_i)}.$$

- C. It is a nonparametric method based on regression trees that estimates IBNR without development triangles.
- D. It minimizes the mean squared prediction error of development factors through numerical optimization.

Answer B

In classical BF, the actuary must *specify the a priori ELR externally* (usually using *pricing* or *industry benchmarks*). In the Cape Cod method (Stanard–Bühlmann), the ELR is *estimated from the portfolio's own data*:

$$\text{ELR}_{\text{CC}} = \frac{\sum_i C_{i,\text{dev.}}}{\sum_i (P_i \times \% \text{ developed}_i)},$$

where

$$\% \text{ developed}_i = \frac{1}{\text{CDF}_i}$$

is the fraction of risk that has already developed, and P_i is the earned premium for year i .

This makes Cape Cod more **empirical and self-consistent**: the ELR is calibrated using the portfolio's observed experience, adjusted for each year's stage of development. It is particularly useful when no reliable a priori estimate is available or when historical *pricing* is not a good guide.

Conceptual comparison:

- *Chain Ladder*: places all weight on emerging experience. Sensitive to volatility in immature years.
- *Bornhuetter–Ferguson*: balances an external a priori estimate against experience. Requires a high-quality ELR.
- *Cape Cod*: estimates the a priori from the data. An intermediate, self-consistent approach, but it can be circular when data are sparse.